

DELE

A Project Synopsis Report Submitted To



SAGE UNIVERSITY, INDORE

Towards Partial Fulfillment for the Award of

**Bachelor of Technology
in
Computer Science & Engineering**

**Submitted By
Arpita Namdev (23ADV3CST1024)**

**Institute of Advance Computing core
SAGE University
Bypass Road, Kailod Kartal, Indore
Madhya Pradesh - 452020
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Abstract

A project inspired from the Airbnb environment implementing the style in to rent everyday gadgets and tools between individuals. Also created a calendar-based booking system preventing double-bookings which also helps renters search for available dates. Integrated secure payments with, hold-release logic protecting both parties from unnecessary issues. An overall review system is added in with a dispute resolution workflow for damaged or missing items.

Introduction

Dele is a web-based marketplace that helps individuals rent out their underutilized everyday gadgets and tools to neighbors and community members. It is inspired by the sharing economy model of Airbnb, it bridges the gap between owners seeking passive income and users that need temporary access to things out of their budget without the burden of maintaining and managing the said product. The system prioritizes trust and security through a booking conflict detection engine, a dual-sided escrow payment system, and a dispute resolution workflow.

Problem Statement

High-quality tools like cameras and generators, sit idle 90% of the time being underutilized while the owners bear the full cost of depreciation. Simultaneously, renters often cannot afford to purchase specialized equipment for one-time use. A web app can solve this with P2P model that enables users to exchange assets directly without intermediaries.

Objective

To develop a digital platform that empowers users to earn a passive income by renting the underutilized everyday gadgets & helps other group of people find the most effective and budget-friendly things for one-time use that won't be a justified purchase. Multiple aspects of the project sets in to make it easier for the user to use and keep it hassle-free.

Key Functional Modules:

1. Lists for owners and discovery option for renters:

- **Dynamic Listings:** Owners create detailed profiles with photos, descriptions, daily rates, and customizable availability calendars for the product.
- **Smart Search:** Users can browse by category (electronics, tools, events) using faceted filters and a real-time availability calendar to prevent booking conflicts.
- **Wishlists & Notifications:** Renters can save items and receive notifications when previously unavailable items become free or when new matching listings appears for them.

2. Bookings & Calendar Logic:

- **Conflict-Free Scheduling:** The backend utilizes **PostgreSQL date range overlap queries** to ensure no double-bookings occur.
- **Buffer Management:** Real-time calendars automatically block off dates for cleaning, maintenance, or owner buffer periods of their products to avoid any issues during checkout.
- **Discounts:** Owners can configure weekly or monthly discount rates to encourage longer-term rentals for certain products.

3. Secure Payment:

- **Hold-and-Release Model:** Integrated with **Razorpay API (for local) and PayPal (for global)**, the system places a payment hold upon booking confirmation. Funds are only released to the owner after the item is returned safely.
- **Security Deposits:** A configurable deposit is automatically held to cover potential damages and is given back after safe return of product to the owner.
- **Damage Claims:** Owners can initiate post-rental damage charges against the renter's saved deposit, supported by a dispute workflow.
- **Earnings Dashboard:** Owners view real-time breakdowns of gross income, platform fees, and net payouts that they receive.

4. Trust, Safety & Dispute Resolution:

- **Identity Verification:** **Stripe Identity** integration is used to verify renters for safety and security. Best for low scale product with its effective price.
- **Two-Sided Feedback System:** Both parties leave star ratings and written feedback after every transaction to build reputation scores as renters and lenders.
- **Dispute Workflow:** Users can submit a claim with photo evidence, which is then sent to an admin for review and resolution.
- **Notifications:** Automated Email and SMS alerts keep users informed of booking requests, confirmations, and return deadlines.

User Experience Goals

- **Mobile Function:** Full functionality for browsing, booking, and managing listings on mobile devices.

- **Frictionless Onboarding:** No complex barriers to entry; quick listing creation and instant booking capabilities.
- **Transparency:** Clear cost previews before confirmation (including deposits).
- **Seamless Transactions:** Smooth money transfer while following hold and release method.

Future Scope

- **Geolocation Filtering:** Auto-detect location and show items within 3-5km first.
- **"Available Now" Badge:** Highlight items that are currently in stock and can be picked up *today*.
- **View Area:** Instead of just a list, offer a map view where users can see pins of available items in their neighborhood.

Language Use and Justification

HTML: Used to create the structure of the webpage, adding in all the important components.

CSS: Helped create and implement design elements to make it interactive and effective for user's experience.

Agile Scrum Methodology

- Why we chose it:
 - Flexibility: Requirements for a rental marketplace (like specific booking rules or user roles) can change as we test the prototype. Agile allows us to adapt quickly.
 - Iterative Delivery: We build the app in small cycles (Sprints) of 2 weeks. First, we build the Login, then the Listing, then the Booking. This ensures we have a working product at every stage.
 - User Feedback: We can show the frontend prototype to users early and fix issues before writing complex backend code.
- The Process:
 1. Sprint Planning: Define what features (e.g., "Calendar UI") will be built in the next 2 weeks.
 2. Daily Stand-ups: Quick checks to track progress and blockers.
 3. Development: Coding the features (HTML/CSS/JS).
 4. Review: Demonstrating the feature to stakeholders.
 5. Retrospective: Discussing what went well and what to improve

Hardware Requirement

(For Developer)

Component	Minimum Requirement	Recommended Requirement
Processor (CPU)	Intel Core i3 (8th Gen) or AMD Ryzen 3	Intel Core i5/i7 (10th Gen+) or AMD Ryzen 5/7
RAM	8 GB	16 GB or higher
Storage	256 GB SSD (Solid State Drive)	512 GB NVMe SSD
Graphics	Integrated Graphics (Intel UHD)	Dedicated GPU (Optional, for UI rendering tests)
Display	1366 x 768 resolution	1920 x 1080 (Full HD) or higher

(For Renters & Owners)

Component	Minimum Requirement	Recommended Requirement
Device	Smartphone (4GB RAM) or Desktop/Laptop	Modern Smartphone (6GB+ RAM) or Tablet/Desktop
Processor	Quad-core (1.5 GHz)	Octa-core (2.0 GHz+)
RAM	4 GB	8 GB
Storage	2 GB free space (for browser cache)	10 GB+ free space
Connectivity	4G / Wi-Fi (10 Mbps)	5G / High-speed Wi-Fi
Camera	5 MP (for ID verification & item photos)	12 MP+ (for high-quality evidence upload)

Software Requirements (Development Tools)

Category	Technology / Tool	Purpose
Operating System	Windows 10/11, macOS, or Linux	Development Environment
Code Editor	VS Code (Visual Studio Code)	Writing and editing HTML, CSS, JS
Frontend Language	HTML5, CSS3, JavaScript (ES6+)	Core logic and UI structure
Version Control	Git & GitHub	Tracking code changes and collaboration
Web Server	Live Server (VS Code Extension) or Node.js	Running the local server for testing
API Testing	Postman or Thunder Client	Testing backend endpoints (future)
Design Tools	Figma or Canva	Prototyping UI/UX before coding